

Which Treatment Should We Choose?

A Bayesian comparison of three treatments

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The question

	Patients	Successes	Rate
A	100	62	0.62
B	100	70	0.70
C	100	66	0.66

Simulated clinical-trial data from the project sheet.

- How large is each success probability, and how sure are we?
- How do the treatments differ?
- What is $P(\text{B better than A})$?
- What is $P(\text{B is the best})$?
- Which treatment should we choose?

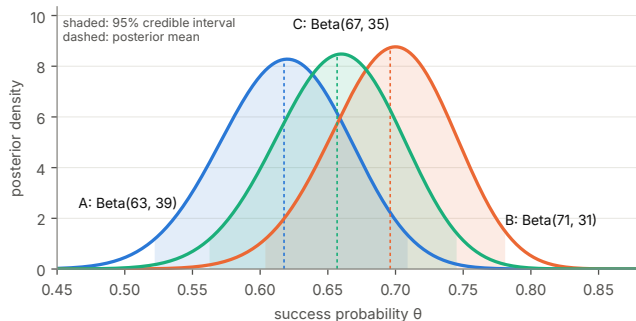
Model: binomial data, Beta prior

- Each patient succeeds independently with probability θ :
 $f(\mathbf{x} \mid \theta) = \theta^s (1 - \theta)^{n-s}$
- Neyman–Fisher: the count s is sufficient; only the table matters
- Uniform prior $\text{Beta}(1, 1)$ gives the conjugate posterior $\text{Beta}(s + 1, n - s + 1)$

$$\theta_A \sim \text{Beta}(63, 39) \quad \theta_B \sim \text{Beta}(71, 31) \quad \theta_C \sim \text{Beta}(67, 35)$$

Module 3 (sufficiency, MLE) • Module 4, Example 4.6 (Beta–binomial)

Estimates and uncertainty

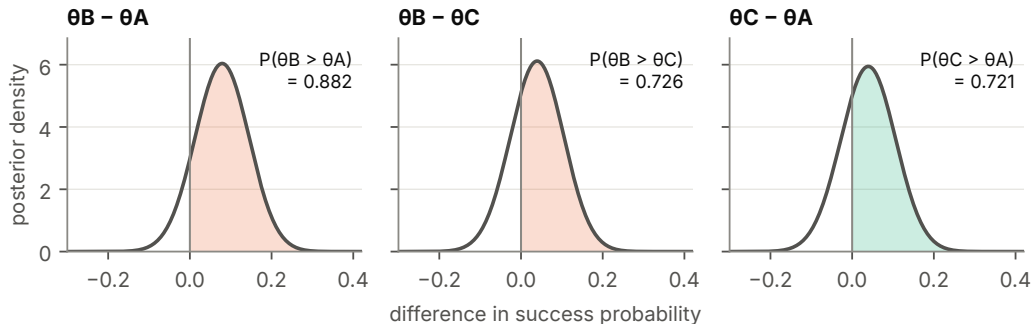


	Post. mean	95% CrI
A	0.6176	0.522–0.709
B	0.6961	0.604–0.781
C	0.6569	0.562–0.745

Each rate is known to about ± 9 points.

Mean < median < mode:
left-skewed posteriors.

How do the treatments differ?



$$P(\theta_B > \theta_A \mid \text{data}) = 0.882$$
$$P(\theta_B - \theta_A > 0.05) = 0.667$$

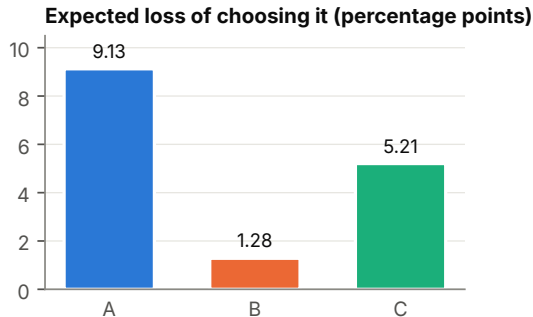
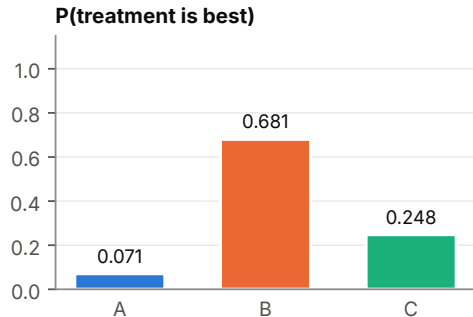
Classical test of B vs A: $p = 0.116$, "not significant".
It cannot say how likely B is to be better.

Exact, not simulated

$$P(\theta_B > \theta_A \mid \text{data}) = \int_0^1 f_B(t) F_A(t) dt$$

- Whole-number Beta parameters: f_B and F_A are **polynomials** with rational coefficients
- So the answer is an exact fraction: 0.882419268177 ...
- Same fraction from an independent finite-sum formula; quadrature agrees to 10^{-12}
- Monte Carlo, 2,000,000 draws: every estimate within 0.71 standard errors

Which treatment is best?



$$P(k \text{ best}) = \int_0^1 f_k(t) \prod_{j \neq k} F_j(t) dt$$

$$P(\text{B best}) = 0.681$$

Choosing is a Bayes decision

Minimise posterior expected loss

Opportunity loss $\max_j \theta_j - \theta_k$

⇒ largest posterior mean: **B**

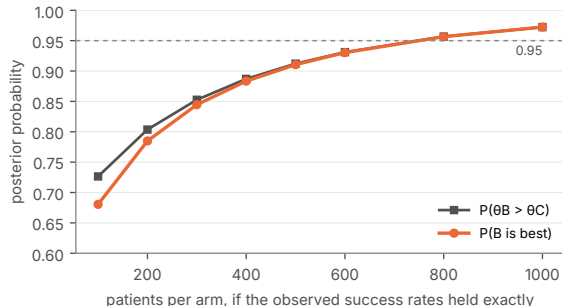
Zero-one loss (not the best)

⇒ largest $P(\text{best})$: **B**

Module 4, Definition 4.3

Prior	$P(\theta_B > \theta_A)$	$P(\text{B best})$	Choice
Beta(1, 1)	0.882	0.681	B
Jeffreys	0.884	0.682	B
Beta(2, 2)	0.880	0.677	B
Beta(10, 10)	0.859	0.652	B

How sure are we about B over C?



- $P(\theta_B > \theta_C) = 0.726$ only
- 95% CrI for $\theta_B - \theta_C$: -0.089 to 0.167
- If the rates held, $P(B \text{ best})$ would reach 0.95 at about 750 patients per arm

Illustration only, not a prediction.

Course check: Lindley and Tierney–Kadane (Module 5)

	Exact $E[\theta]$	Lindley error	T-K error	Exact $E[\text{odds}]$	T-K error
A	0.617647	-4.7×10^{-5}	-5.2×10^{-5}	1.657895	-6.3×10^{-5}
B	0.696078	-7.8×10^{-5}	-6.2×10^{-5}	2.366667	-1.9×10^{-4}
C	0.656863	-6.3×10^{-5}	-5.7×10^{-5}	1.970588	-1.1×10^{-4}

- Our code reproduces the slides' example: Lindley 1.9965, T-K 1.995138
- Errors $\approx 10^{-4} = O(n^{-2})$; Lindley is *exact* for the odds, $(s+1)/(n-s)$

Recommendation

Choose treatment **B**

- Highest success rate: 0.6961 (95% CrI 0.604–0.781)
- Most likely best: 0.681; smallest expected loss: 1.28 points
- The same under every prior we tried

Caveats: B vs C is not settled; simulated data; success is the only outcome; independent arms assumed. A further B-versus-C trial is justified if the stakes are high.